T028 bracket learning report

Project: T028 bracket tutorial
Analysis: T028 bracket tutorial · frozen revision 5
Run completed: 2026-10-16T10:39:16.928405+00:00
Report generated: 2026-10-16T10:39:35.025302+00:00
Display units: mm-N-MPa

Report f33bbdef-9ef6-45a2-914f-2d42ddf565f0
Run 7bd07c58-53c4-48ac-9d31-1cca2f96592b
Input SHA-256 9d9051d721db9394c303248886cce08fe07b983d480ab376e405f6e452de1ff7

Scope, warnings and assumptions

> Gravity unavailable: density is missing.

> Yield factor of safety unavailable: yield strength is missing.

> Same undeformed physical coordinates and explicit element/body/material sides in both retained affine C3D10 meshes. Quadratic displacement; tensor interpolation precedes invariants, with the declared unaveraged/partitioned-smoothed mode. Changes are compared minus reference; relative change is absolute change / absolute reference, unavailable at or below the disclosed SI reference floor. Floors are display conventions, not numerical acceptance tolerances. Raw native extrema are separate moving locations. One arbitrary comparison cannot establish convergence, accuracy, stability or safety.

> The raw peak magnitude increased at its mesh-dependent location. This may indicate an unresolved concentration or singularity; it does not prove divergence or convergence. Inspect the common physical samples separately.

stress_singularity · all bodies · unassessed: Peaks near supports, reentrant corners or idealized load/bond edges may increase with refinement. No automatic singularity detection is claimed. Compare fixed physical locations away from these regions; a lower smoothed peak is not evidence of convergence or a design allowable.

yielding · 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1 · unassessed: Compare raw integration-point von Mises with the frozen supplied ductile yield. Yield exceedance does not model plastic deformation; missing yield cannot establish a factor. No fatigue, brittle failure, buckling or code check is implied.

small_deformation · 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1 · not triggered: Maximum native displacement / largest body extent is a 1% screening heuristic. Exceedance requires review of small-deformation linear elasticity. Passing this heuristic does not bound local strain/rotation or certify validity; display exaggeration is irrelevant.

stress_singularity · all bodies · unassessed: Peaks near supports, reentrant corners or idealized load/bond edges may increase with refinement. No automatic singularity detection is claimed. Compare fixed physical locations away from these regions; a lower smoothed peak is not evidence of convergence or a design allowable.

yielding · 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1 · unassessed: Compare raw integration-point von Mises with the frozen supplied ductile yield. Yield exceedance does not model plastic deformation; missing yield cannot establish a factor. No fatigue, brittle failure, buckling or code check is implied.

small_deformation · 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1 · not triggered: Maximum native displacement / largest body extent is a 1% screening heuristic. Exceedance requires review of small-deformation linear elasticity. Passing this heuristic does not bound local strain/rotation or certify validity; display exaggeration is irrelevant.

Linear static, small-deformation and supplied material assumptions apply. No automatic design-safety conclusion is generated.

Yield assessment does not assess fatigue, brittle failure, buckling or plasticity. Stress singularities require engineering review.

Passing equilibrium does not establish convergence, stability or physical validity. Target-laptop validation and remaining external references are pending; bounded M3 approval does not certify this design.

Figures are light report derivatives of retained verified data. Hidden bodies, clipping, smoothing and deformation do not alter all-body native extrema or warnings.

Selected figures and setup

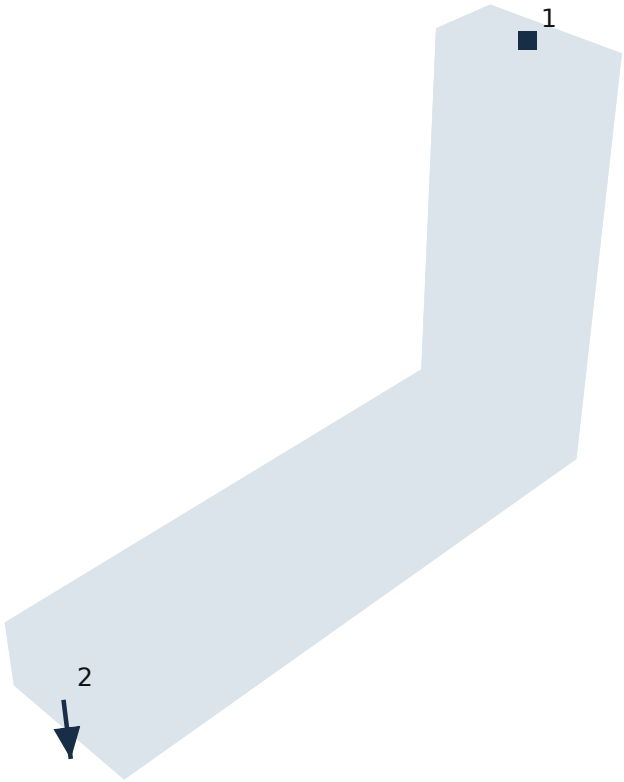
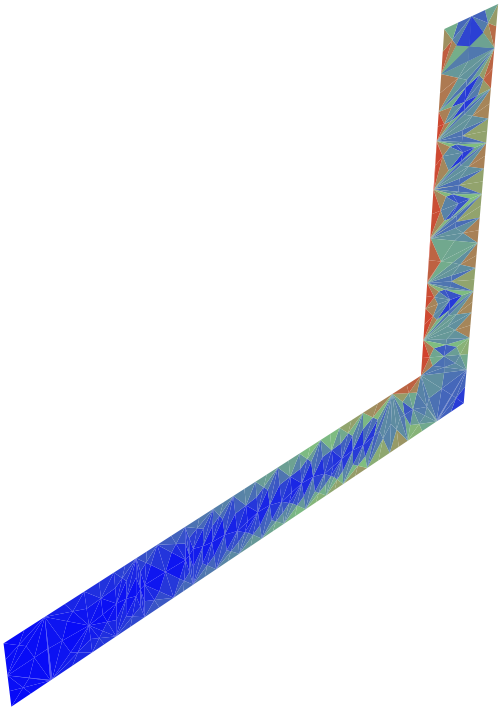


Figure 1. Frozen loads and supports on undeformed geometry. Arrows indicate global direction only; squares indicate fixed supports. Marker numbers resolve to the setup table; arrow lengths do not encode magnitude.



Von Mises stress (MPa): 0.0214198 to 2.56102

Colour limits clamp display only; raw extrema remain independent.

Figure 2. Von Mises stress; global Cartesian; element-interpolated volume section {'point_m': [0.0, 0.01, 0.0], 'normal': [0.0, 1.0, 0.0]}; linear_C3D10_extrapolation_from_native_DAT_IP; element-owned tensors, then invariants; default unaveraged, optional body/material-partitioned smoothing. Deformation ×0; visible bodies 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1. Flat triangle colour samples use the vertex-scalar mean; no numerical extrema are read from pixels. Camera and SI limits are retained in the report model.

Geometry and source units

Fact	Frozen value
Geometry revision	c22b8127-da06-4000-87a4-33fa52eb3c71
Source length unit	m
Bounds (mm)	[0, 0, 0, 80, 20, 60]
Dimensions (mm)	[80, 20, 60]
Bodies / faces	1 / 8
Geometry SHA-256	2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae

Material properties

Body / material	E (MPa) / Poisson ratio	Density (tonne/mm³)	Yield (MPa)
2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1 / Synthetic bracket reference	210000 / 0.3	Unavailable	Unavailable

Material provenance

Body	Detail	Source / assumption
2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1	Grade / condition	Synthetic / Test only

Body	Detail	Source / assumption
2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1	Temperature / assumptions	20 C reference / Synthetic planar bracket, E=210 GPa, nu=0.3. No certified allowable.
2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1	Library / original units	None / None / mm-N-MPa
2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1	Independent synthetic axial fixture	No URL supplied; accessed not supplied

Loads and supports

Assignment	Immutable targets	Physical value / direction
Fixed root / fixed	2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1:f5	All translations fixed
Axial pressure / force	2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1:f2	Total force N: [0, 0, -10]

Setup figure markers

Marker	Assignment / kind	Position (mm) / global direction
1	28476ccc-4681-45c3-875e-7f13a0eff42e / fixed	[5, 10, 60] / [0, 0, 0]
2	58850b3c-c6d3-4e74-b105-1028695d1c6e / force	[80, 10, 5] / [0, 0, -1]

Bond review and gaps

Connection	Master / slave	Approved gap / review
None	No bonded connections	Single body or disconnected setup as frozen

Mesh and solver

Fact	Frozen value
Element / node count	C3D10 / 501 elements / 1122 nodes
Quality	{"schema_version": 1, "accepted": true, "minimum_jacobian_m3": 1.1388222501047388e-07, "minimum_mean_ratio": 0.41470101292932926, "bad_elements": [], "warning_elements": [], "all_affine": true, "policy": {"jacobian_floor_m3": 1e-15, "mean_ratio_floor": 0.05, "mean_ratio_warning": 0.2, "subdivision_depth": 3, "method": "cubic tetrahedral Bernstein
	certificate with outward rounding margin"}}
Mesh controls	{"schema_version": 1, "geometry_revision_id": "c22b8127-da06-4000-87a4-33fa52eb3c71", "global_size_m": 0.008, "source_global_size": 8.0, "source_units": "mm-N-MPa", "local_sizes": [], "curvature_elements": 0, "recovery_enabled": false}
Solver	CalculiX 2.23
Solver binary SHA-256	d55622f40e0435e738a86cbd364d525fff31524212ecca17d71d4c15ea2935f5
Original solve CPU/RAM/thread inventory	Not recorded in this legacy local run; resource policy and measured usage are retained in run artifacts.
Native build SHA-256	917a915f2aa78656131eeba825c13dece6cf81834d786a317d7dc9e6ef6db74d
Numerical method	Linear static / small deformation / global Cartesian / C3D10; no stabilization

Raw native extrema — all bodies

Field / location	Minimum	Maximum / identity
Displacement magnitude (mm); node	0; node; node 9; element None; IP None; body None; material None; position [10, 20, 60] mm	0.01277876; node; node 40; element None; IP None; body None; material None; position [80, 10, 0] mm
Von Mises stress (MPa); integration_point	0.05241925; integration_point; node None; element 427; IP 2; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [4.961612, 10.69098, 58.40424] mm	2.22156; integration_point; node None; element 795; IP 3; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [9.329311, 15.44766, 11.0192] mm

Field / location	Minimum	Maximum / identity
Displacement ux (mm); node	-0.004128333; node; node 41; element None; IP None; body None; material None; position [80, 16.66667, 0] mm	3.818768e-06; node; node 758; element None; IP None; body None; material None; position [4.984894, 20, 57.51676] mm
Displacement uy (mm); node	-3.26033e-05; node; node 113; element None; IP None; body None; material None; position [10, 20, 31.42857] mm	3.246152e-05; node; node 131; element None; IP None; body None; material None; position [10, 0, 31.42857] mm
Displacement uz (mm); node	-0.01209357; node; node 40; element None; IP None; body None; material None; position [80, 10, 0] mm	0.0005512615; node; node 15; element None; IP None; body None; material None; position [0, 3.333333, 0] mm
Stress σxx (MPa); integration_point	-1.735528; integration_point; node None; element 765; IP 3; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [13.20875, 15.79867, 0.7191769] mm	2.210785; integration_point; node None; element 797; IP 4; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [11.13764, 4.54495, 9.337211] mm
Stress σyy (MPa); integration_point	-0.7432398; integration_point; node None; element 785; IP 4; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [0.690983, 8.509288, 59.20212] mm	0.7863091; integration_point; node None; element 772; IP 4; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [9.309017, 8.509288, 59.20212] mm
Stress σzz (MPa); integration_point	-2.01368; integration_point; node None; element 785; IP 4; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [0.690983, 8.509288, 59.20212] mm	2.476119; integration_point; node None; element 795; IP 3; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [9.329311, 15.44766, 11.0192] mm
Stress σxy (MPa); integration_point	-0.1918412; integration_point; node None; element 636; IP 2; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [7.374908, 15.2373, 9.999019] mm	0.1791619; integration_point; node None; element 586; IP 3; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [7.374908, 4.708555, 9.847929] mm
Stress σyz (MPa); integration_point	-0.2778591; integration_point; node None; element 829; IP 2; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [0.6525951, 17.73633, 59.31223] mm	0.2682029; integration_point; node None; element 811; IP 2; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [9.270629, 17.7381, 59.33192] mm
Stress σzx (MPa); integration_point	-0.6747697; integration_point; node None; element 795; IP 3; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [9.329311, 15.44766, 11.0192] mm	0.7419845; integration_point; node None; element 775; IP 3; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [4.140497, 0.8058208, 7.597365] mm
Principal stress σ1 (MPa); integration_point	-0.6341068; integration_point; node None; element 785; IP 4; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [0.690983, 8.509288, 59.20212] mm	2.708633; integration_point; node None; element 795; IP 3; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [9.329311, 15.44766, 11.0192] mm
Principal stress σ2 (MPa); integration_point	-0.7432099; integration_point; node None; element 785; IP 4; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [0.690983, 8.509288, 59.20212] mm	1.076399; integration_point; node None; element 732; IP 4; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [8.756874, 3.902735, 9.762426] mm
Principal stress σ3 (MPa); integration_point	-2.050081; integration_point; node None; element 785; IP 4; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [0.690983, 8.509288, 59.20212] mm	0.6670126; integration_point; node None; element 772; IP 4; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [9.309017, 8.509288, 59.20212] mm

Yield assessment — frozen supplied strength / raw integration-point von Mises

Body / material	Yield (MPa)	Factor / critical location
2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1 / Synthetic bracket reference	Unavailable	Unavailable — missing yield; integration_point; node None; element 795; IP 3; body 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1; material M1; position [9.329311, 15.44766, 11.0192] mm

External reactions about global origin

Body / constrained nodes / DOFs	Force (N)	Moment (N·mm)
2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1 / 39 / 117	[-8.9e-07, 1.03e-06, 10]	[100, -800.0001, 1.770862e-06]

Force and moment equilibrium

Balance	Residual	Reference	Criterion / limit	Outcome
force (N)	[-8.9e-07, 1.03e-06, 4.393e-06]; norm 4.59907e-06	10	relative; relative 4.59907e-07; allowed 0.05	Passed
moment (N·mm)	[4.736667e-05, -9.2215e-05, 1.770862e-06]; norm 0.0001036839	806.2258	relative; relative 1.28604e-07; allowed 4.031129	Passed

Equilibrium method and actions

Fact	Value
Method	Frozen physical load magnitudes plus nonzero prescribed-component actuator reactions; moments about global origin; external constrained span only; internal ties/MPC/free-node RF excluded
Reaction method	DAT RF minus written equivalent CLOAD/pressure/consistent gravity, constrained global components only; unique nodes/DOFs per body; internal ties excluded
Moment origin (m)	[0, 0, 0]
Applied force (N)	[0, 0, -10]
Applied moment (N·mm)	[-100, 800, 0]
Stress / extrema method	linear_C3D10_extrapolation_from_native_DAT_IP; element-owned tensors, then invariants; default unaveraged, optional body/material-partitioned smoothing; Displacement: all native DAT nodes; von Mises/yield factor: all raw DAT integration points, including interior

Physical refinement comparison — no convergence certification

Sample / body / position (m)	Reference / compared (MPa)	Signed / absolute change	Relative change
Sample 1 / 2ce744e6cafd0f6a3c1c18eb022b035a3da8c274ea70f5411f440d02bc8d97ae:b1 / [0.04, 0.01, 0.0075]	0.6206693 / 0.6033576	-0.01731173 / 0.01731173	0.02789204

Comparison provenance and moving raw peaks

Fact	Value
Compared run	8f8b04c6-b8bc-440d-ba6b-f4f7d55c89e2
Compared result SHA-256	d6a941807f3bc342d1abad69d6556794a1df66dbfa87767a68e02b7356afe454
Physics SHA-256	6ed59d355640f922e7792c45e32b2f6dc6076be6f43bbfbf007925c9438a9d27
Method / denominator floor	Same undeformed physical coordinates and explicit element/body/material sides in both retained affine C3D10 meshes. Quadratic displacement; tensor interpolation precedes invariants, with the declared unaveraged/partitioned-smoothed mode. Changes are compared minus reference; relative change is absolute change / absolute reference, unavailable at or
	below the disclosed SI reference floor. Floors are display conventions, not numerical acceptance tolerances. Raw native extrema are separate moving locations. One arbitrary comparison cannot establish convergence, accuracy, stability or safety.; 1e-06 Pa
Reference / compared raw peaks	2.22156 / 2.565148
Raw peak semantics	Native peaks may move between meshes; physical samples and moving extrema remain separate.

User-written conclusion

Attributed to the report author; this text is not a computed engineering finding.

Synthetic isotropic small-deformation learning exercise. Physical point [0.04, 0.01, 0.0075] m; von_mises: [{"reference":620669.3382313023,"compared":603357.6042785969,"signed_change":-17311.7339527054,"absolute_change":17311.7339527054,"re
Surface peaks require separate convergence review.

Reproducibility IDs

schema_version: 1

job_id: 7bd07c58-53c4-48ac-9d31-1cca2f96592b

analysis_id: a4463908-637f-4b37-ade1-d4e51aaf30b0

analysis_revision: 5

geometry_revision_id: c22b8127-da06-4000-87a4-33fa52eb3c71

input_sha256: 9d9051d721db9394c303248886cce08fe07b983d480ab376e405f6e452de1ff7

results_sha256: d85d60b4adf0ac965debd0130c2d57b6262b0c73e018826621a8ba6a42e190a6

mesh_sha256: f689a96248d44ef3c8ea5698f7efb3017579b7ceed58f19b58558f14f316c678

deck_sha256: 71624bcdbe3bcb9cb8bcf413efa3f27f2b3f139ddece4047d0101cbdc9b49794

dat_sha256: e554d330959c32ace5baeab685903b3476f3e09fa54d3661b50dcf8f197e5cab

services/worker/worker/postprocess/view.py: 7ea99bb6c7c9daf7277201e6779148eba316d236c0699a9629348e8900124a8c

services/worker/worker/postprocess/inspection.py: 2671965a0e9881217553f96e8389289d24f389b7b002569b370fe92a01486f08

services/worker/worker/postprocess/comparison.py: 8970cbe6b7e9686a77176bc4d8ae432217276bf4f2f57577c55c837941d8a3aa

services/api/app/domain/comparisons.py: 7031bcf9a45c54cef209210382718a5057449d41e29fded6bab882316e389bc0

services/api/app/domain/refinement.py: b63ca772fc862a035dda9f2c3f63d06326fa3cbda50f093e5efae4b4c258a996

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services/worker/worker/postprocess/tensors.py: 9dc5efb68ebfafc05d5f964019548b801fdcad8706032554334aa0c537767584

services/worker/worker/meshing.py: 1d71f8f57bd020f96804f284f4be1d2a0fd51bf89d0bc529f48cd1aad2a52e9e

services/worker/worker/actions.py: 1a8edff017a1397b45fa667d9056a2b5e9ebf250248812928855f64dec460e9b

services/worker/worker/directions.py: 8de17d5705b81b651a5077ddf3b80e3b29a13b9f453af1c79fc2e82db06e4592

services/worker/worker/calculix/results.py: da43f91763e9880a6e8728906202f1394d7b89a31af61e6a6f163c1190c29609

services/api/app/jobs/service.py: 15988d904ab64da1ef736c9708982926d4970ce042a72bb2668e0809b739f18d

services/api/app/domain/results.py: c4c0b9b18b090bea8352ea59a2a09776d3b20783598f81c0db6b6ce18706b3ad

version: report-v1

source_commit: 743e1f3fbf8f2b01a00195466f9ccb754fdcd304

source_dirty: true

generator_python: 3.12.14

generator_platform: Linux-6.18.44-x86_64-with-glibc2.41

generator_cpu_count: 3

font_dejavu_regular_sha256: 57f73e11f51999432bf7ab22ce55b6f945d5eca1bf824404cfa9ec2e3718c84e

chromium_binary_sha256: 4fa61f0ba0b4f89b7bec242c32ca16d3bccf69dbfef0651924a9ba8fede5f9d6

node_binary_sha256: bc17c508ffeed0ec622934f9b7fa72f8e78da65350e63c3eceb56fa688aa5e12

services/api/app/domain/reports.py: 45a4b8349f9ae22f3bb63d50f7ae8acce9e8129136a6969f2cbdfd8274789ca0

services/api/app/domain/report_service.py: 1e417c01670abd940a3d73f22e6555d96095dbfa88db782c6efc6380b42cd776

services/worker/worker/reports/model.py: adcb4b46695ab4ddad618402cd8d6bb613332fa0a2c3ddd97d6227d0caf7dd44

services/worker/worker/reports/figures.py: bedc9575b4b72309897e2eba8fc0664ac04e84b7de74615e1d595e29eca19047

services/worker/worker/reports/html.py: a266e66c2204f726528b7bc307935b2ef7096153fbbc58146ee5d7a7c1316bb9

services/worker/worker/reports/content.py: 5914c8af6cddf62125566746a8d881fcdb5a2ef8d40a84bc77095e248e929b3e

services/worker/worker/reports/render.py: 8b4f6ae521693827346fe2ab67de5e680cc583cdcdbf4d2780c6f70db0dd98de

scripts/reports/pdf.mjs: 125def705cc32a64a6ace989b65a573dac6af6de54fd02c1ccd11064ea65c123

package-lock.json: feb1470637caf3d65d9af3d08e89ca777ca097f49b055845452b610917c45fba

uv.lock: 36f59cf0e56b6442588403ba9ac6b47e732a46ce4a9d8cd7f3e5edcea741e261

Download the immutable report-model JSON for complete original inputs, view definitions, camera, quality and tool policy. Its SHA-256 and this PDF’s SHA-256 are in the report record.